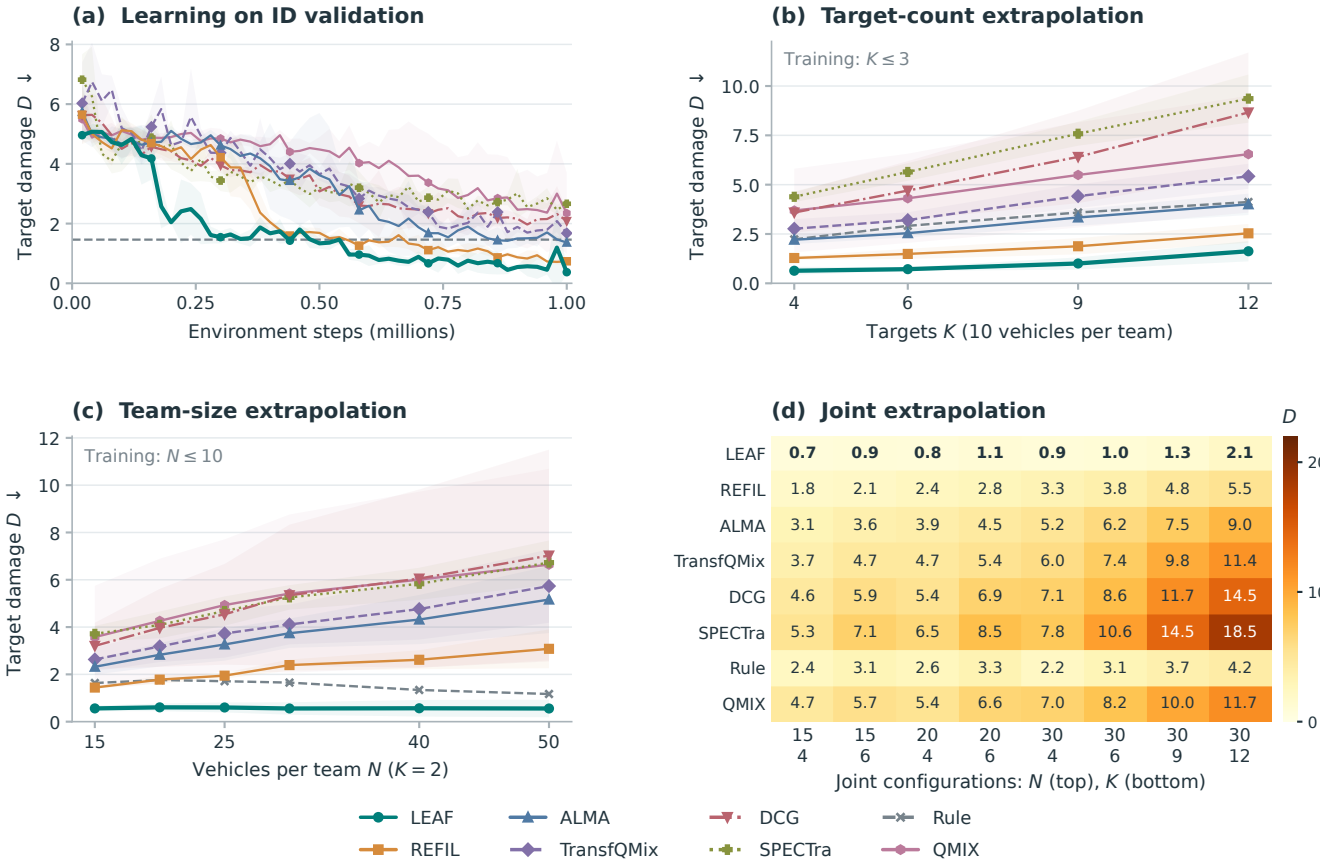
(a) Learning on ID validation
Target damage D ↓
Environment steps (millions)
(b) Target-count extrapolation
Training: K ≤ 3
Target damage D ↓
Targets K (10 vehicles per team)
(c) Team-size extrapolation
Training: N ≤ 10
Target damage D ↓
Vehicles per team N (K = 2)
(d) Joint extrapolation
D
LEAF 0.7 0.9 0.8 1.1 0.9 1.0 1.3 2.1
REFIL 1.8 2.1 2.4 2.8 3.3 3.8 4.8 5.5
ALMA 3.1 3.6 3.9 4.5 5.2 6.2 7.5 9.0
TransfQMix 3.7 4.7 4.7 5.4 6.0 7.4 9.8 11.4
DCG 4.6 5.9 5.4 6.9 7.1 8.6 11.7 14.5
SPECTra 5.3 7.1 6.5 8.5 7.8 10.6 14.5 18.5
Rule 2.4 3.1 2.6 3.3 2.2 3.1 3.7 4.2
QMIX 4.7 5.7 5.4 6.6 7.0 8.2 10.0 11.7
15 15 20 20 30 30 30 30
4 6 4 6 4 6 9 12
Joint configurations: N (top), K (bottom)
LEAF
REFIL
ALMA
TransfQMix
DCG
SPECTra
Rule
QMIX